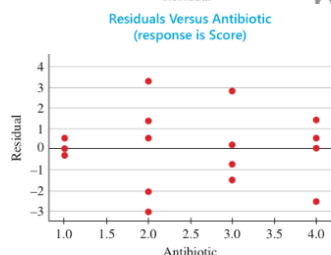
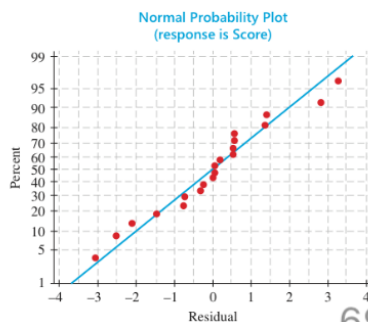


Good Tasting Medicine A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.⁷ The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

Child	Antibiotic			
	1	2	3	4
1	4.8	2.2	6.8	4.6
2	8.1	9.2	6.6	9.6
3	5.0	2.6	3.6	6.5
4	7.9	9.4	5.3	8.5
5	3.9	7.4	2.1	2.0

- a. What design is used in collecting these data?
b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?



- c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

a. randomized block design

b.

① data in each treatment should follow normal distribution ✓

② have common variance. X

data follow the normal distribution

but. Antibiotic 1 has small variance and Antibiotic 2's variance is big, they don't have common distribution

c.

H_0 : The four distributions are the same.

H_a : At least one distribution is different in location.

$$T_1 = 12, T_2 = 13, T_3 = 10, T_4 = 15, k = 4, b = 5$$

$$\alpha = 0.05, df = 4 - 1 = 3$$

$$F_r = \frac{12}{5 \cdot 4 \cdot 5} \sum T_i^2 - 3b(k+1) \sim \chi^2_{0.05, 3}$$

$$F_{r*} = \frac{12}{5 \cdot 4 \cdot 5} (12^2 + 13^2 + 10^2 + 15^2) - 3 \cdot 5 \cdot 5 = 1.56$$

$$RR: \{F_r \mid F_r > 7.815\} \quad 1.56 < 7.815 \Rightarrow \text{non-reject } H_0$$

$$\chi^2_{0.05, 3} = 7.815$$

There's no sufficient evidence to say that there is at least one distribution is different in location.